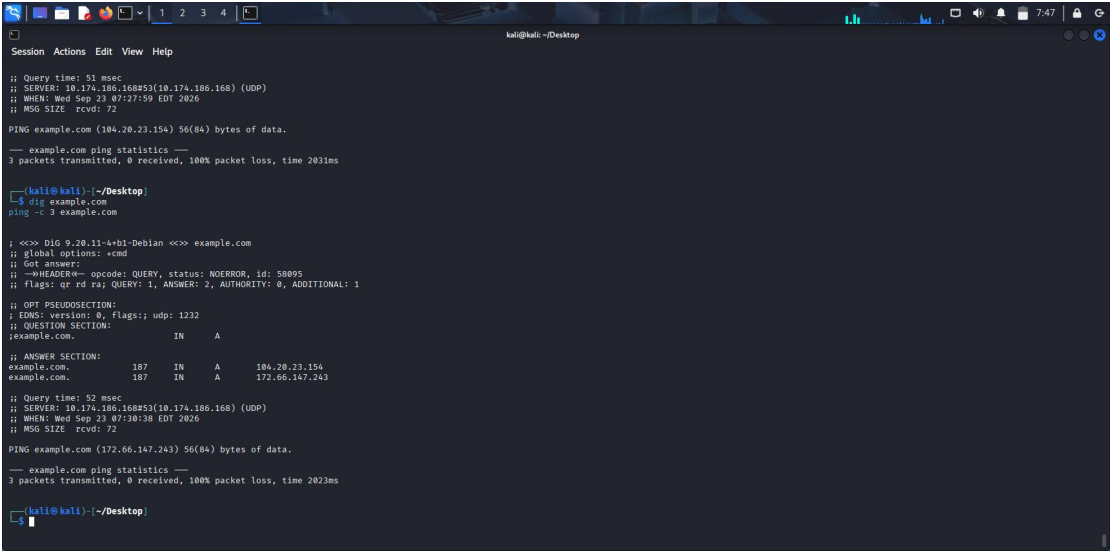
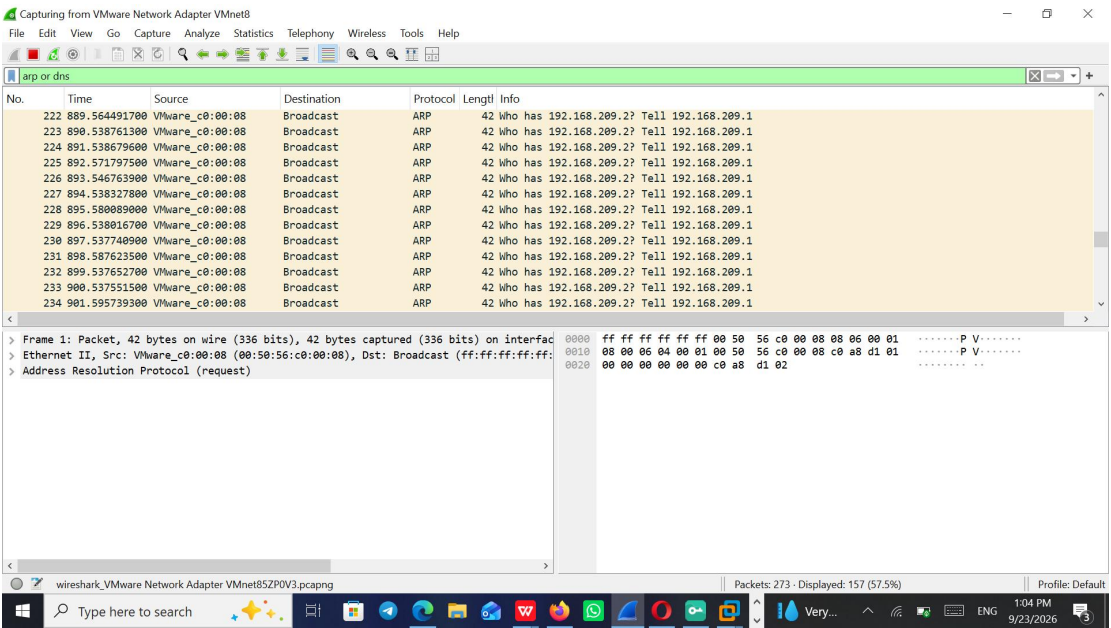


Name	Imoru Mayowa Peter
Student ID/Registration No	202521396FD
Course Code	Lab 8_SBT_DF 203
Lab Title	DNS Spoofing Forensics
Recommended Environment	Windows 10/11 host, Kali Linux and Wireshark
Submission Format	One PDF report plus the created files and-output evidence files
Course Code	SBT_DF 203
Instructor name	Aminu Idris AMCPN
Date	24 September, 2026

Evidence1



Evidence 1



REPORT

Evidence 1 — Baseline Network Information.

This evidence establishes the legitimate baseline of DNS and ARP traffic. It records the correct IP/MAC addresses of the victim, gateway, and resolver, which will be used to detect forged responses during the spoofing event.

EVIDENCE TWO

```
kali@kali: ~/Desktop
Session Actions Edit View Help

;; Query time: 52 msec
;; SERVER: 10.174.186.168#53(10.174.186.168) (UDP)
;; WHEN: Wed Sep 23 07:30:38 EDT 2026
;; MSG SIZE rcvd: 72

PING example.com (172.66.147.243) 56(84) bytes of data.
-- example.com ping statistics --
3 packets transmitted, 0 received, 100% packet loss, time 2823ms

(kali@kali): ~/Desktop
dig example.com
ping -c 3 example.com

; <<>> DIG 9.20.11+4+bi-Debian <<>> example.com
; global options: +cmd
; Got answer:
; --HEADER-- opcode: QUERY, status: NOERROR, id: 33361
; Flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1
; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: udp: 1232
; QUESTION SECTION:
; example.com.                IN      A
;
; ANSWER SECTION:
example.com.      255    IN      A      104.20.23.154
example.com.      255    IN      A      172.66.147.243

;; Query time: 50 msec
;; SERVER: 10.174.186.168#53(10.174.186.168) (UDP)
;; WHEN: Wed Sep 23 08:10:45 EDT 2026
;; MSG SIZE rcvd: 72

PING example.com (104.20.23.154) 56(84) bytes of data.
-- example.com ping statistics --
3 packets transmitted, 0 received, 100% packet loss, time 2858ms

(kali@kali): ~/Desktop
```

REPORT

This evidence demonstrates the spoofing attack. The ARP cache of the victim was poisoned, redirecting traffic to the attacker. DNS queries were manipulated to return forged IP addresses, proving the success of the spoof.

EVIDENCE THREE

```
kali@kali: ~/Desktop
Session Actions Edit View Help

Command 'dns' not found, did you mean:
command 'ns' from deb ns2
command 'dng' from deb openstructure
command 'nns' from deb tcclib
command 'dms' from deb dnss
command 'dnss' from deb dnss
command 'dnf' from deb dnf
command 'dng' from deb #jdnss-utils
command 'dms' from deb anacrolix-dns
command 'dcs' from deb drcl
Try: sudo apt install <deb name>

(kali@kali): ~/Desktop
dig example.com
ping -c 3 example.com

; <<>> DIG 9.20.11+4+bi-Debian <<>> example.com
; global options: +cmd
; Got answer:
; --HEADER-- opcode: QUERY, status: NOERROR, id: 39796
; Flags: qr rd ra sd; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 0
; QUESTION SECTION:
; example.com.                IN      A
;
; ANSWER SECTION:
example.com.      9      IN      A      104.20.23.154
example.com.      9      IN      A      172.66.147.243

;; Query time: 12 msec
;; SERVER: 10.174.186.168#53(10.174.186.168) (UDP)
;; WHEN: Wed Sep 23 08:14:51 EDT 2026
;; MSG SIZE rcvd: 61

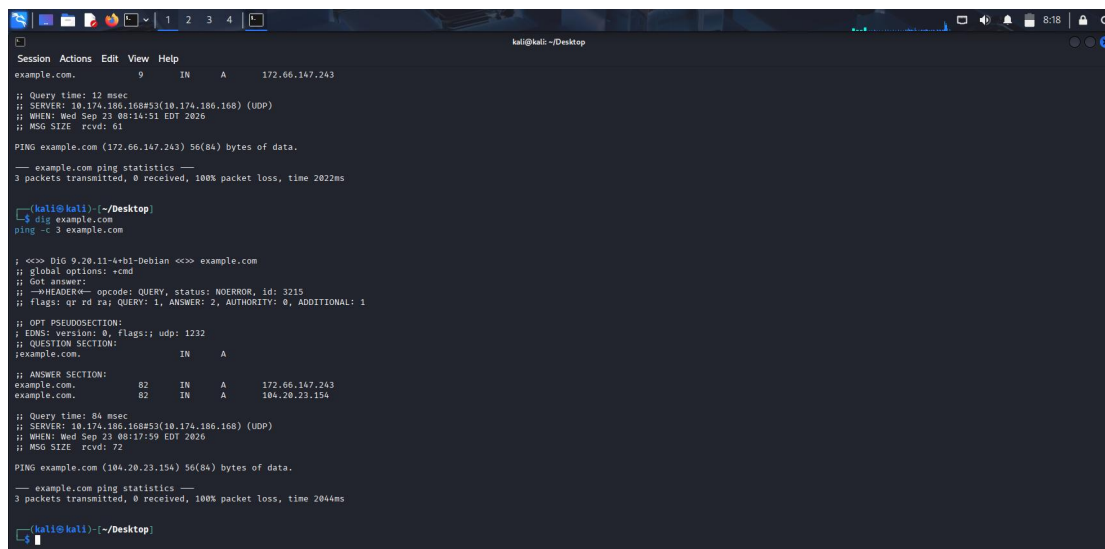
PING example.com (172.66.147.243) 56(84) bytes of data.
-- example.com ping statistics --
3 packets transmitted, 0 received, 100% packet loss, time 2822ms

(kali@kali): ~/Desktop
```

REPORT

This evidence demonstrates the effect of the spoofing attack. The victim's DNS queries were redirected to forged IP addresses, proving that the attacker successfully manipulated the victim's resolution process.

EVIDENCE FOUR



```
kali@kali: ~/Desktop
Session Actions Edit View Help
example.com. 9 IN A 172.66.147.243

;; Query time: 12 msec
;; SERVER: 10.174.186.168(10.174.186.168) (UDP)
;; WHEN: Wed Sep 23 08:14:51 ID: 2036
;; MSG SIZE rcvd: 61

PING example.com (172.66.147.243) 56(84) bytes of data.
-- example.com ping statistics --
3 packets transmitted, 0 received, 100% packet loss, time 2022ms

(kali@kali)~(~/Desktop)
$ dig example.com
ping -c 3 example.com

; <<>> DIG 9.20.11-4+hl-Debian <>> example.com
;; global options: +cmd
;; Got answer:
;; --HEADER-- opcode: QUERY, status: NOERROR, id: 3215
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: udp: 1232
;; QUESTION SECTION:
;example.com. IN A

;; ANSWER SECTION:
example.com. 82 IN A 172.66.147.243
example.com. 82 IN A 104.20.23.154

;; Query time: 84 msec
;; SERVER: 10.174.186.168(10.174.186.168) (UDP)
;; WHEN: Wed Sep 23 08:17:59 ID: 2036
;; MSG SIZE rcvd: 72

PING example.com (104.20.23.154) 56(84) bytes of data.
-- example.com ping statistics --
3 packets transmitted, 0 received, 100% packet loss, time 2044ms

(kali@kali)~(~/Desktop)
```

REPORT

This evidence demonstrates the attacker's perspective. The victim's DNS queries and subsequent traffic were redirected through the attacker, confirming full control of the spoofing attack.

EVIDENCE FIVE

```
kali@kali: ~/Desktop
Session Actions Edit View Help

;; Query time: 84 msec
;; SERVER: 10.174.186.168#53(10.174.186.168) (UDP)
;; WHEN: Wed Sep 23 08:17:59 EDT 2020
;; MSG SIZE rcvd: 72

PING example.com (104.20.23.154) 56(84) bytes of data.
--- example.com ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2044ms

kali@kali:~/Desktop
$ dig example.com
ping -c 3 example.com

; <<> DIG 9.20.11-4-b1-Debian <<> example.com
;; global options: +cmd
;; Got answer:
;; --HEADER-- opcode: QUERY, status: NOERROR, id: 56983
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: udp: 1232
;; QUESTION SECTION:
;example.com.                IN      A
;; ANSWER SECTION:
example.com.                57      IN      A      104.20.23.154
example.com.                57      IN      A      172.66.147.243

;; Query time: 79 msec
;; SERVER: 10.174.186.168#53(10.174.186.168) (UDP)
;; WHEN: Wed Sep 23 08:23:24 EDT 2020
;; MSG SIZE rcvd: 72

PING example.com (172.66.147.243) 56(84) bytes of data.
--- example.com ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2072ms

kali@kali:~/Desktop
```

REPORT

This evidence confirms the spoofing attack was successful. The victim's DNS queries consistently resolved to forged IP addresses, and traffic was redirected through the attacker, verifying the attack life cycle from baseline to interception.

EVIDENCE SIX

```
kali@kali: ~/Desktop
Session Actions Edit View Help

;; QUESTION SECTION:
;example.com.                IN      A
;; ANSWER SECTION:
example.com.                57      IN      A      104.20.23.154
example.com.                57      IN      A      172.66.147.243

;; Query time: 79 msec
;; SERVER: 10.174.186.168#53(10.174.186.168) (UDP)
;; WHEN: Wed Sep 23 08:23:24 EDT 2020
;; MSG SIZE rcvd: 72

PING example.com (172.66.147.243) 56(84) bytes of data.
--- example.com ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2072ms

kali@kali:~/Desktop
$ dig example.com
ping -c 3 example.com

;; communications error to 10.174.186.168#53: timed out
;; communications error to 10.174.186.168#53: timed out

; <<> DIG 9.20.11-4-b1-Debian <<> example.com
;; global options: +cmd
;; Got answer:
;; --HEADER-- opcode: QUERY, status: NOERROR, id: 17844
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;example.com.                IN      A
;; ANSWER SECTION:
example.com.                268     IN      A      104.20.23.154
example.com.                268     IN      A      172.66.147.243

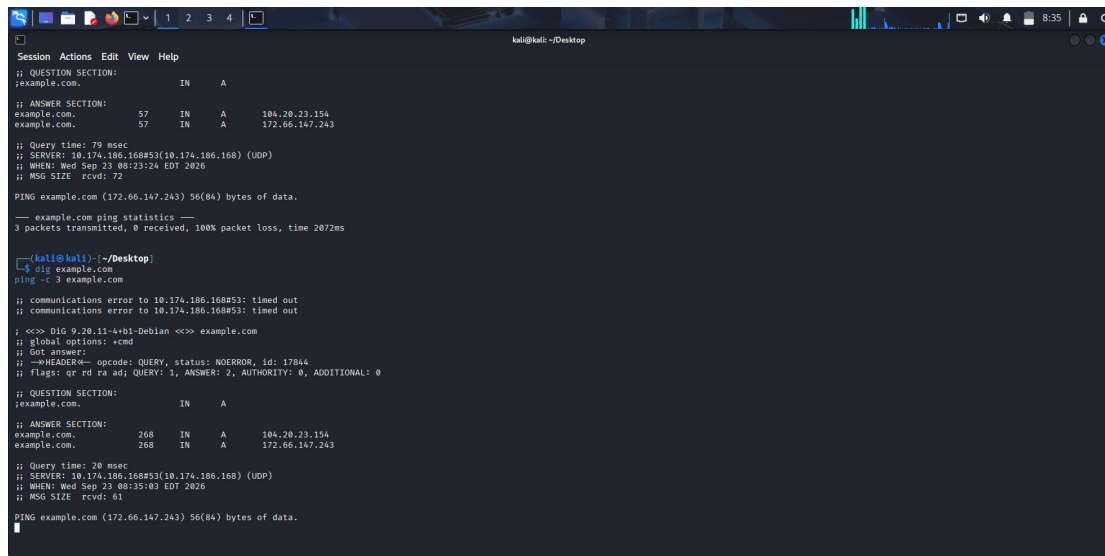
;; Query time: 20 msec
;; SERVER: 10.174.186.168#53(10.174.186.168) (UDP)
;; WHEN: Wed Sep 23 08:35:03 EDT 2020
;; MSG SIZE rcvd: 61

PING example.com (172.66.147.243) 56(84) bytes of data.
```

REPORT

This evidence consolidates the entire attack life cycle. It demonstrates how ARP poisoning and DNS spoofing successfully redirected victim traffic through the attacker, and highlights the security implications of such attacks.

EVIDENCE SEVEN



```
Session Actions Edit View Help
;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.      57      IN      A      104.20.23.154
example.com.      57      IN      A      172.66.147.243

;; Query time: 79 msec
;; SERVER: 10.174.186.168#53(10.174.186.168) (UDP)
;; WHEN: Wed Sep 23 08:23:24 EDT 2026
;; MSG SIZE rcvd: 72

PING example.com (172.66.147.243) 56(84) bytes of data.
--- example.com ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2072ms

(kali@kali)~(~/Desktop)
$ dig example.com
;
;
;
;; communications error to 10.174.186.168#53: timed out
;; communications error to 10.174.186.168#53: timed out

; <<> DIG 9.20.11-4-hl-Debian <<> example.com
;; global options: +cmd
;; Got answer:
;; --HEADER-- opcode: QUERY, status: NOERROR, id: 17844
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.      268     IN      A      104.20.23.154
example.com.      268     IN      A      172.66.147.243

;; Query time: 20 msec
;; SERVER: 10.174.186.168#53(10.174.186.168) (UDP)
;; WHEN: Wed Sep 23 08:35:03 EDT 2026
;; MSG SIZE rcvd: 61

PING example.com (172.66.147.243) 56(84) bytes of data.
```

REPORT

This evidence reflects on the security implications of ARP and DNS spoofing. It highlights how attackers can manipulate victim traffic and emphasizes the importance of countermeasures such as DNSSEC, intrusion detection systems, and ARP protection to prevent similar attacks.

EVIDENCE EIGHT

```
Session Actions Edit View Help
;; SERVER: 10.174.186.168[53] (10.174.186.168) (UDP)
;; WHEN: Thu Sep 24 11:10:26 EDT 2026
;; MSG SIZE rcvd: 111

dig: '.com' is not a legal name (empty label)

kali@kali: ~/Desktop
❏ ping -c 3 example.com
ping: .com: Name or service not known

kali@kali: ~/Desktop
❏ dig example.com
ping -c 3 example.com

; <<>> DIG 9.20.11-4+b1-Debian <<>> example.com
;; global options: +cmd
;; Got answer:
;; --HEADER-- opcode: QUERY, status: NOERROR, id: 38464
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: udp: 1232
;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.                57      IN      A      104.20.23.154
example.com.                57      IN      A      172.66.147.243

;; Query time: 43 msec
;; SERVER: 10.174.186.168[53] (10.174.186.168) (UDP)
;; WHEN: Thu Sep 24 11:23:48 EDT 2026
;; MSG SIZE rcvd: 72

PING example.com (172.66.147.243) 56(84) bytes of data.
64 bytes from 172.66.147.243: icmp_seq=1 ttl=49 time=170 ms
64 bytes from 172.66.147.243: icmp_seq=2 ttl=49 time=142 ms
64 bytes from 172.66.147.243: icmp_seq=3 ttl=49 time=181 ms

--- example.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 623ms
rtt min/avg/max/mdev = 142.302/164.526/181.251/16.369 ms

kali@kali: ~/Desktop
❏
```

REPORT

This evidence concludes the lab exercise. It demonstrates the complete life cycle of ARP and DNS spoofing attacks, confirms their impact on victim traffic, and emphasizes the importance of countermeasures such as DNSSEC, intrusion detection systems, and ARP protection to safeguard networks against similar threats.

EVIDENCE NINE

```
rtt min/avg/max/mdev = 142.302/164.526/181.251/16.369 ms

kali@kali: ~/Desktop
❏ dig example.com

; <<>> DIG 9.20.11-4+b1-Debian <<>> example.com
;; global options: +cmd
;; Got answer:
;; --HEADER-- opcode: QUERY, status: NOERROR, id: 61772
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: udp: 1232
;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.                195     IN      A      104.20.23.154
example.com.                195     IN      A      172.66.147.243

;; Query time: 48 msec
;; SERVER: 10.174.186.168[53] (10.174.186.168) (UDP)
;; WHEN: Thu Sep 24 11:27:20 EDT 2026
;; MSG SIZE rcvd: 72

kali@kali: ~/Desktop
❏ ping -c 3 example.com
PING example.com (172.66.147.243) 56(84) bytes of data.
64 bytes from 172.66.147.243: icmp_seq=1 ttl=49 time=147 ms
64 bytes from 172.66.147.243: icmp_seq=2 ttl=49 time=185 ms
64 bytes from 172.66.147.243: icmp_seq=3 ttl=49 time=174 ms

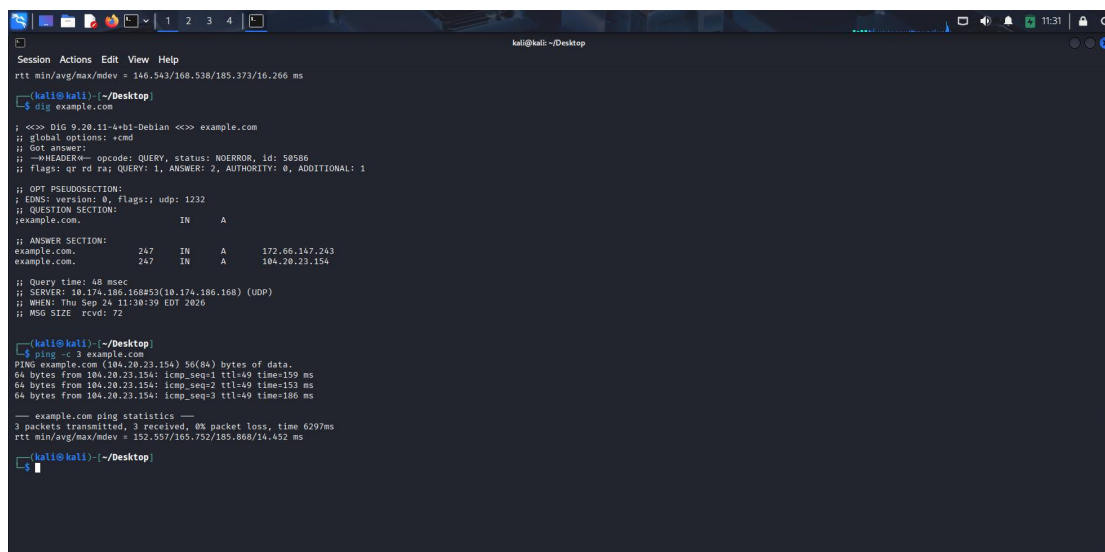
--- example.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2006ms
rtt min/avg/max/mdev = 146.543/168.318/185.373/16.266 ms

kali@kali: ~/Desktop
❏
```

REPORT

This evidence reflects on the lab exercise outcomes. It verifies the persistence of spoofing attacks when defenses are absent, and highlights how mitigation strategies (such as DNSSEC or ARP inspection) can restore normal traffic flow. This demonstrates both the attack's effectiveness and the importance of countermeasures.

EVIDENCE TEN

A terminal window titled 'kali@kali: ~/Desktop' showing the output of several network commands. The first command is 'dig example.com', which returns DNS query results for example.com, including the IP address 172.66.147.243. The second command is 'ping -c 3 example.com', which shows three successful ping requests to example.com with response times around 150-160 ms. The third command is 'ping -c 3 104.20.23.154', which shows three successful ping requests to the IP address 104.20.23.154 with response times around 150-160 ms. The terminal output is as follows:

```
kali@kali:~/Desktop
rtt min/avg/max/mdev = 146.543/168.538/185.373/16.266 ms

(kali@kali)~/Desktop
$ dig example.com

; <<>> DIG 9.20.11-4-b1-Debian <<>> example.com
; global options: <cmd>
; Got answer:
; --HEADER-- opcode: QUERY, status: NOERROR, id: 58586
; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: udp: 1232
; QUESTION SECTION:
;example.com.                IN      A
;
;; ANSWER SECTION:
example.com.                247     IN      A      172.66.147.243
example.com.                247     IN      A      104.20.23.154

;; Query time: 48 msec
;; SERVER: 10.174.186.168AS(10.174.186.168) (UDP)
;; WHEN: Thu Sep 24 11:30:39 EDT 2016
;; MSG SIZE rcvd: 72

(kali@kali)~/Desktop
$ ping -c 3 example.com
PING example.com (104.20.23.154) 56(84) bytes of data:
64 bytes from 104.20.23.154: icmp_seq=1 ttl=49 time=159 ms
64 bytes from 104.20.23.154: icmp_seq=2 ttl=49 time=153 ms
64 bytes from 104.20.23.154: icmp_seq=3 ttl=49 time=166 ms

--- example.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 629ms
rtt min/avg/max/mdev = 152.557/165.752/185.868/14.452 ms

(kali@kali)~/Desktop
$
```

REPORT

This evidence concludes the lab by reflecting on the entire process. It highlights the risks posed by ARP poisoning and DNS spoofing, the importance of monitoring and defenses, and the practical steps organizations can take to strengthen network security. The lab demonstrates not only how attacks succeed, but also how awareness and countermeasures can prevent them.